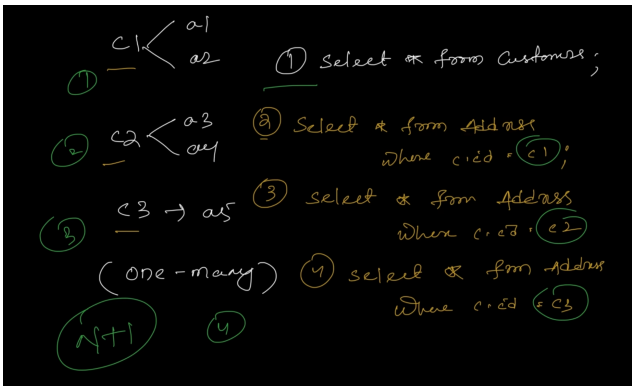
N+1 Fix - https://github.com/Java-Techie-jt/entitygraph-jpa/tree/main

Customer Vs Address

One query to fetch all customers and n queries to fetch all their addresses



customer 1 | Output

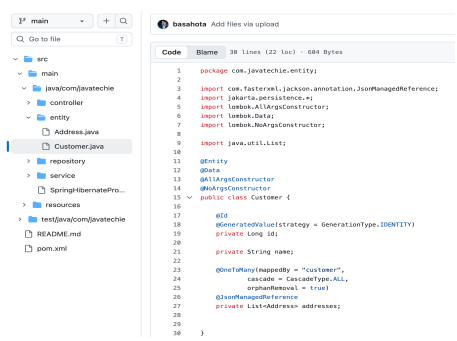
SELECT * FROM Customer c

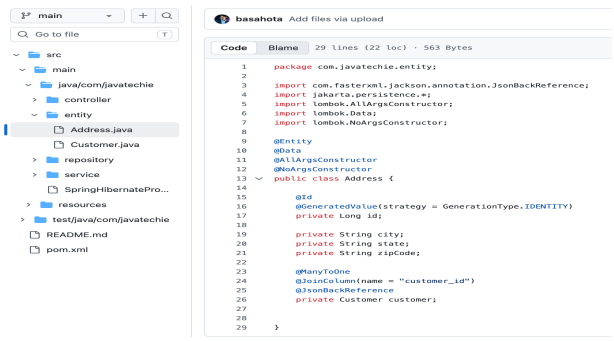
	id	name
1	1	John
2	2	Bob
3	3	Charlie

address 1 | Output

SELECT * FROM Address a

	id	city	state	zipCode	customer_id
1	1	New York	NY	10001	1
2	2	Los Angeles	CA	90001	1
3	3	Chicago	IL	60601	2
4	4	Houston	TX	77001	2
5	5	San Francisco	CA	94101	3
6	6	Miami	FL	33101	3
7	7	Seattle	WA	98101	3






```
public List<Customer> getAllCustomers() { 1 usage
    return customerRepository.findAll();
}
```

N+1 queries in console

```
Run  SpringHibernatePropsApplication
2025-04-06T15:39:13.606+05:30  INFO 11718 --- [spring-hibernate-props] [nio-9191-exec-1] o.a.c.c.C
2025-04-06T15:39:13.607+05:30  INFO 11718 --- [spring-hibernate-props] [nio-9191-exec-1] o.s.web.s
2025-04-06T15:39:13.608+05:30  INFO 11718 --- [spring-hibernate-props] [nio-9191-exec-1] o.s.web.s
Hibernate: select c1_0.id,c1_0.name from Customer c1_0
Hibernate: select a1_0.customer_id,a1_0.id,a1_0.city,a1_0.state,a1_0.zipCode from Address a1_0 whe
Hibernate: select a1_0.customer_id,a1_0.id,a1_0.city,a1_0.state,a1_0.zipCode from Address a1_0 whe
Hibernate: select a1_0.customer_id,a1_0.id,a1_0.city,a1_0.state,a1_0.zipCode from Address a1_0 whe
```

Implement these two methods. You call either one you notice only one query in the console.

```
Address.java
Customer.java
repository
AddressRepository...
CustomerRepositor...
service
CustomerService.j...
```

```
8   import java.util.List;
9
10
11  public interface CustomerRepository extends JpaRepository<Customer, Long> {
12
13      @EntityGraph(attributePaths = "addresses")
14      List<Customer> findAll();
15
16      @Query("SELECT c FROM Customer c LEFT JOIN FETCH c.addresses")
17      List<Customer> fetchCustomersWithAddress();
18  }
```

```
Run  SpringHibernatePropsApplication
2025-04-06T15:49:46.829+05:30  INFO 13362 --- [spring-hibernate-props] [nio-9191-exec-1] o.a.c.c.C
2025-04-06T15:49:46.829+05:30  INFO 13362 --- [spring-hibernate-props] [nio-9191-exec-1] o.s.web.s
2025-04-06T15:49:46.831+05:30  INFO 13362 --- [spring-hibernate-props] [nio-9191-exec-1] o.s.web.s
Hibernate: select c1_0.id,a1_0.customer_id,a1_0.id,a1_0.city,a1_0.state,a1_0.zipCode,c1_0.name fro
```